Pixels, Frames & Transformers

Text in Images and Multimodal Posts · Module 11

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Meaning in the Combination

Much of what we study is neither pure text nor pure image:

- a **meme**, where the caption and the picture comment on each other
- a **cereal box**, with health claims in print next to a cartoon mascot
- a **post**, where a few words change how a photo reads

Two questions for this module

1. How well can we get the text *out* of an image?
2. What does each modality contribute to the meaning, and does a model use both?

Getting the Text Out: OCR or a VLM?

Classic OCR (Tesseract)

finds lines of text and recognizes them letter by letter

+ fast, and only reports what it reads
− struggles with logos, curved text, low contrast, small print

A vision-language model

sees the whole image and *writes* the text it believes is there

+ reads stylized brand names and odd layouts
− slow, and it can "complete" text that isn't printed

A yardstick without hand-coding: the product name in the Open Food Facts database is almost always printed on the front, so we measure the share of its words each reader finds.

When the Reader Writes

Words that only the VLM reports are a mix of

- text that Tesseract missed (logos, curved slogans), and
- text the model **supplied**: a familiar slogan of the brand, a typical claim ("rich in fiber"), the full name when only part is visible

You can only tell them apart by looking. Check a sample by eye before you analyze VLM transcriptions, and report how often invented text occurs.

Name recall can't catch invented text: it only asks whether the expected words are there.

Health Claims: Two Imperfect Benchmarks

The VLM codes four claims on the front of each box: whole grain, fiber, less sugar, organic. Open Food Facts records many claims as **labels**, entered by contributors.

VLM yes, database no: often a real claim never entered as a label, sometimes a claim invented for a box that just *looks* healthy.

Database yes, VLM no: often a claim printed on the side or back, which a front photo doesn't show, or a small logo missed at low resolution.

Neither source is ground truth for "what the front of the box claims". For a study of that construct, hand-code a sample of it.

A Controlled Meme Experiment

SUPPORTIVE CAPTION	MOCKING CAPTION
the same photo	the same photo

six protest photos × two captions = 12 memes

Within each pair the photo is identical, so **only the caption** can tell the two versions apart. That makes it possible to see what a model actually uses.

Adding a caption adapts a Commons photo. CC BY and CC BY-SA allow that; if you share the result, credit the author and keep the share-alike license.

Four Ways to Read a Meme

	what the model sees	what to expect
1	the caption text only	does well: the stance is in the words
2	the photo only	exactly 50% by design: both versions share the photo
3	CLIP on the meme	as good as CLIP's ability to read the caption
4	the VLM on the meme	as good as the VLM's reading and reasoning

Then turn it around: one cheerful caption ("What a lovely day in the city") on King's Day photos, where it is sincere, and on protest photos, where it reads as sarcastic. Does the model use the picture to read the caption? Small models often don't.

What Does Each Modality Add?

An **ablation**: the same classifier on

- text only (OCR of the package)
- image only (CLIP embedding)
- both

Outcome: favorable Nutri-Score (A, B) versus unfavorable (D, E). With about a hundred products, use cross-validation, not one split.

Report the spread. Across five random fold splits, scores typically move by a few points, so small differences between versions mean little.

"Both" doesn't always win: more features give a small sample more ways to overfit.

Designs for Multimodal Posts

1. **Code the combined meaning**, and separately the text alone and the image alone; the difference is what the image adds
2. **Ablate** with the same model: text only, image only, both
3. **Swap** captions between images to see which modality a model relies on
4. **Measure text extraction** before treating text in images as data
5. **Check access conditions**: many meme and post datasets come with their own license (Hateful Memes, Kiela et al., 2020, uses Meta's), and platforms have terms for collecting new posts

In the Notebook

- Reading packaging with Tesseract and with Qwen3-VL, scored by product-name recall
- Health claims coded by the VLM versus database labels
- A controlled meme experiment: caption, photo, CLIP, VLM
- Same caption, different photo: sincere or sarcastic?
- Text only, image only, both: an ablation with cross-validation

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11_text_in_images_and_multimodal_posts_exercises.ipynb · 5 exercises · GPU needed